

5 (a) Nerves consist of bundles of different types of neurones.

Fig. 5.1 shows the mean transmission speed of impulses for three types of sensory neurone, **C**, **A^δ** and **A^β**.

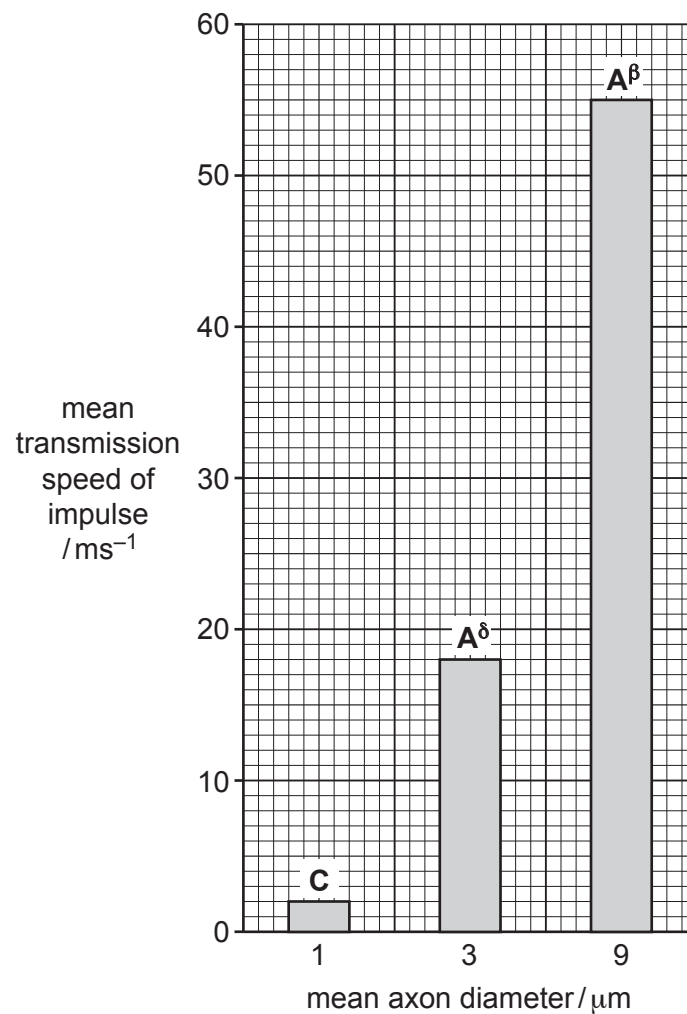


Fig. 5.1

- Impulses from pain receptors in the skin are sent along sensory neurones **C** and **A^δ**.
- Sensory neurones **C** and **A^δ** synapse in the spinal cord with relay neurones known as projection neurones.
- Projection neurones send impulses to the part of the brain that perceives pain.
- Impulses from touch receptors in the skin pass along sensory neurones **A^β**, which can also synapse with the projection neurones in the spinal cord.
- Sensory neurones **A^δ** and **A^β** are myelinated.
- Sensory neurones **C** are unmyelinated.

(i) Describe the results shown in Fig. 5.1 **and** suggest explanations for these results.

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(ii) Gently rubbing a damaged area of skin can reduce the perception of pain.

Suggest an explanation for this.

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- (b) Back pain in humans can be reduced by using a transcutaneous electrical nerve stimulation (TENS) machine.

A TENS machine uses electrical impulses to stimulate nerve endings near the site of the pain.

It is thought that TENS triggers the release of natural painkillers called endorphins at synapses.

Fig. 5.2 shows synapses in a pain pathway.

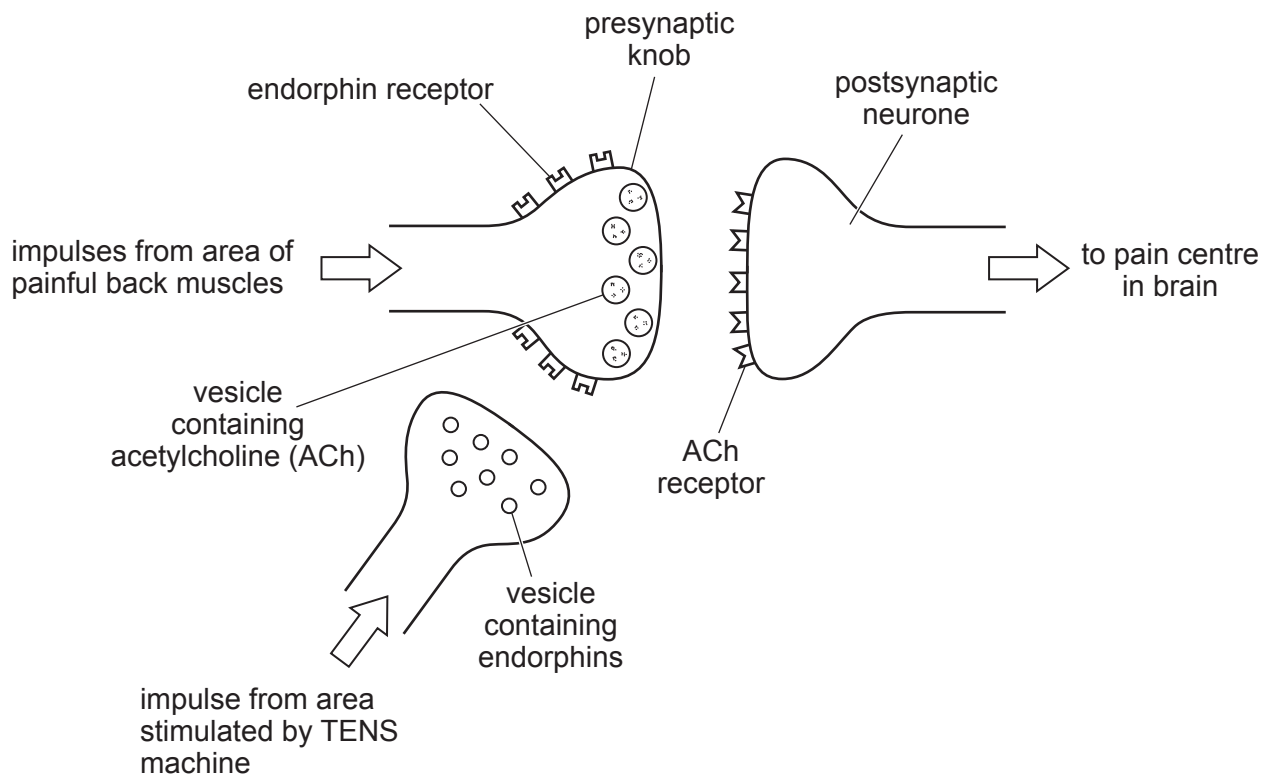


Fig. 5.2

(i) Suggest **and** explain how endorphins may act to reduce pain.

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(ii) Suggest **one disadvantage** of using pharmaceutical drugs for reducing pain compared to using TENS.

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